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Adding structure to the transition process to advanced mathematical activity

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The transition process to advanced mathematical thinking is experienced as traumatic by many students. Experiences that students had of school mathematics differ greatly to what is expected from them at university. Success in school mathematics meant application of different methods to get an answer. Students are not familiar with logical deductive reasoning, required in advanced mathematics. It is necessary to assist students in this transition process, in moving from general to mathematical thinking. In this article some structure is suggested for this transition period. This essay is an argumentative exposition supported by personal experience and international literature. This makes this study theoretical rather than empirical.

Keywords: advanced mathematical thinking; concept image and concept definition; natural and formal learners; communication in mathematics; logical thinking

1. Introduction

The transition process to advanced mathematical thinking e.g. from high school to university or from calculus to analysis is experienced as traumatic by many students. The actual processes of 'doing problems' and 'proving statements' are essentially very different and require a significant change in thinking style. The ways of working in advanced mathematics differ greatly from what was expected of the students at school level. Most students' perception of being successful in mathematics in school does not involve much inquiry but mostly just the application of different methods. To make the transition to advanced mathematics, students need to learn how to work formally with definitions and theorems. Applying deductive reasoning is essential and few students possess this skill naturally.

Tall [1] described the transition:

> The move from elementary to advanced mathematical thinking involves a significant transition: that from describing to defining, from convincing to proving in a logical manner based on those definitions.... It is the transition from the coherence of elementary mathematics to the consequences of advanced mathematics, based on abstract entities which the individual must construct through deductions from formal definitions. (p. 20)

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Marais [2] relates a seminar held at the Mathematics Institute of the University of Warwick on how students learn. Students from each year group were asked to give a short talk about how they experienced their year as mathematics students. The first year student opened his talk by writing in big letters on the blackboard: '*FEEL THE FEAR OF MATHEMATICS*...' and continued to talk about how difficult it was for them to make the transition from school to university mathematics.

In many senior level undergraduate mathematics classrooms, students are introduced in a formal way to mathematical theories with little chance of constructing concepts or building meaning through experiences. It might be because of this presentation that some students fail to realize the importance of developing the mental skills or tools they need to really understand mathematics. They see mathematics as a system of algorithms to be performed to get 'the right' answer.

Finding explanations for the magnitude of this jump has been the object of many research studies. According to Alcock and Simpson [3] there is a significant difference between *mathematical thinking* and *general thinking*. The study of Alcock and Simpson [3] is one of a number of studies that seem to indicate that mathematical thinking (including deductive reasoning) is not something that comes naturally, but is a skill that has to be acquired by most students entering the field of advanced mathematics.

Students in university mathematics courses are typically confronted at first with a few basic definitions of terms central to the development of the vocabulary of the course. From their experience at school, however, it is likely that they do not view these definitions as important, but still think that the essence of the course is in the problems and all they will have to do is to learn how to respond correctly when confronted with various situations. A study by Nelan [4] showed that students in their study of mathematics see very little relationship between vocabulary and techniques and do not believe that they have to understand the meaning of words in order to succeed in mathematics. Students still think in terms of stimulus and response and see no need for independent thinking. Nelan pointed out that students come to a mathematics class with a certain amount of intellectual baggage concerning the nature of mathematics. Trying to teach mathematics to students who expect something quite different is a hard task.

Mathematics is recorded and communicated through logical statements – definitions, theorems, conjectures and proofs. Along with other components, such as creativity and intuition, logic therefore forms an integral part of mathematical thinking; however, even adults sometimes find it difficult to reason in a formal logical way, since propositional logic is not a good model for human reasoning. To help students reason mathematically, we need to teach them the skills they do not possess naturally.

2. Mathematical thinking and general thinking

Students arriving at university have not really been exposed to thinking mathematically. They are used to *general thinking* in which the principal way of arriving at general interpretations is through empirical induction. In general thinking, facts are normally accumulated to convince us of the validity of a particular statement. Validation depends on the quantity and variety of confirmatory facts that we accumulate [5].

In contrast to dealing with the general world, a large part of mathematics is based on an axiomatic system in which deductive reasoning is the only accepted route to gain new mathematical knowledge. Concepts are defined and theorems are proved using the rules of logic. The concept of a mathematical proof is outside this normal way of thinking, for it offers an absolute guarantee of the truth of a mathematical statement. The idea that no more evidence is required is in conflict with our normal search for additional confirmation [5].

To be fair, empirical induction is also used in mathematics to arrive at conjectures and the role of discovery, explanation and analytical methods which provide elements that mathematicians employ in the route to a proof is equally important, but actual proof depends highly on logical deduction.

The first time many students make contact with the axiomatic world of mathematics is at university when they are asked to prove mathematical statements concerning concepts that are precisely defined.

Fischbein [5] believes that the student's mind must undergo a fundamental modification in order to understand what a mathematical proof really means.

In contrast with other sciences, the mathematician's laboratory is mainly confined to his/her mind. Mathematics involves the creation of a new mental world in the mind that may be entirely hypothetical [6]. Mathematicians develop a personal world of concepts and relationships related to his/her field of study that includes ideas of what ought to be true before necessarily being able to prove it.

> The entire investigative process of the mathematician may be performed mentally, in accordance with a certain axiomatic system . . . [7, p. 149].

3. Concept image and concept definition

A distinction between *concept image* and *concept definition* was made by Tall and Vinner [8] based on the fact that many concepts in mathematics have been encountered in some informal way before they are formally defined in advanced mathematics courses. This cognitive structure of a mathematical concept is called the concept image, while the formal words used to specify the concept is called the concept definition.

Tall and Vinner [8] showed that students do not always only use definitions of mathematical concepts to make decisions in mathematics. Students build their mental models according to the concept image, which can be different from the concept definition, and sometimes make mistakes in reasoning. Mathematics students have to develop the skill to understand when something is a description of a concept and when is it a formal definition. Students should develop the ability to use both the concept image and the formal concept definition when doing mathematics [3].

Students in general find mathematical syntax hard to absorb. Marais [2] found that one of the reasons why students struggled with using definitions and proof was that they did not possess the ability to use the basic syntax of mathematics. I recall an incident where I was explaining the idea of an accumulation point of a set to a group of second year students. After my first informal explanation it was clear that the majority of the students had a very good idea of what I meant. Then I gave them the formal definition. The students immediately lost their enthusiasm and it was clear

that I had lost them. One student even asked 'Are we expected to be able to do these translations?'

4. Natural and formal learners

Based on a distinction made by Duffin and Simpson [9], two different kinds of mathematics students are identified, the *natural* learner and the *formal* learner [2,10,11].

In this classification, for a *natural learner* the new knowledge to be learnt fits the individual's internal framework. For a *formal learner* the new experience has no connections with previous experiences. The learner either has to change the structures or form a new structure since his/her internal structures want to account for the situation in different ways.

> ...the natural learner may attempt to give meaning to the definitions from personal experience, giving meaning to the theory, whilst the formal learner attempts to gain meaning from the definitions, extracting meaning from the theory [11, p. 19].

For natural learners, new advanced concepts in mathematics present fewer problems – these new experiences are not in conflict with their previous experiences. Reality, however, is that many of the students that we have are formal learners. For formal learners, learning how to use the rules of logic can be accomplished by rote procedures, fixed rules; for example interchanging quantifiers when negating statements. Students coming from high school are normally more familiar with this form of learning and it may give them a sense of confidence in mathematics that they seem to lose when confronted with the magnitude of new concepts that they are exposed to at university.

Should we accept the idea that many students do not think logically naturally, but use mental models to make decisions, it is understandable that mathematics students may find it difficult to cope with the expectation that they should behave logically within the axiomatic system of mathematics. In advanced mathematics, the focus should not only be on understanding the concept, but also on the formal mathematical syntax.

5. Logic in mathematics

The first treatise on logic, consisting of a collection of rules for deductive reasoning was written by Aristotle. It was intended to serve as a basis for the study of every branch of knowledge. The English mathematicians Boole and de Morgan realized that one of the major problems in studying logic was that their only tool was everyday language and they created an algebra of statements, thereby founding modern symbolic logic.

Dubinsky and Yiparaki [12] claim that predicate calculus can play an important role in learning mathematics as a tool for making sense of and analysing mathematical statements that are not otherwise understood. Concepts in mathematics, such as *uniform continuity* are difficult to understand, even for the more able students, and as students move into more advanced mathematics, they are confronted with more and more unfamiliar concepts that are more difficult to imagine or visualize. In many instances formal logical structure may be the only tool available to students to deal with the situation in an attempt to make sense of the mathematics. If logic is taught as a tool in the beginning of undergraduate

courses, even natural learners can develop the skill to use logical structure to cope with complex mathematical situations [2].

Selden and Selden [13] investigated undergraduate students' ability to write informal mathematical statements into the language of formal predicate calculus. They found that students could not reliably determine the logical structure of common mathematical statements. They claim that the students in their study would not be able to determine the correctness of mathematical proofs. In their study they focused on the procedural understanding or mental skills that students need to develop to be able to validate proofs. These skills, including the unpacking of the logic of a mathematical statement, according to them, do not come naturally but have to be taught to most students.

Two major components of logic that feature prominently in mathematics are *quantification* and *negation*. Fluent handling of quantification and negation can assist in making students' concept definitions and concept images compatible. Understanding of quantification is necessary to understand most concepts in mathematics, for example, linear independence, compactness, uniform continuity, etc. Concepts like these and others in mathematics require students to be able to express a quantified proposition in formal language, to negate the statement and to reason about the original statement and its negation [14]. At a conceptual level, when complex logical statements are used to define a concept, it may be said that in order to understand what something is, it is essential to understand what it is not [14].

Dubinsky *et al*. [14] identified three methods of negation:

- Negation by rules: expressing the statement in formal language and applying rules such as de Morgan's laws.
- Negation by meaning: using mental representations of the meaning of the statement and taking the complementary set of situations to be the negation.
- Negation by recursion: the statement is 'parsed' so that it appears to be a single operation between two statements. The 'top level' statement is first negated after which the lower levels follow. Negation is done either by rules or by meaning.

This distinction is supported by Marais who, in her study with first year students at the University of Warwick, distinguished between negation by expressing the statement in formal language and applying rules, and negation using mental representations of the meaning of the statement and taking the complementary set of situations to be the negation [2]. The first is used by formal learners and the second more by natural learners.

Research on logic in the field of mathematics education focuses on students' difficulty in understanding concepts in logic and their inability to apply the rules of logic in other fields of mathematics. Many concepts in mathematics require students to be able to express a quantified proposition in formal language, to negate the statement and to reason about the original statement and its negation [14].

6. Communicating mathematics

Mastering undergraduate mathematics is sometimes considered to be a two-step process. First, students have to understand the mathematical concepts [15,16] and

second, they have to be able to communicate their understanding of these concepts in written format [17].

In the first step, the lecturer clarifies concepts by using two verbal languages: a commonly spoken, everyday language and a subject-specific, scientific language. Therefore, the student has to be proficient in both these languages [18]. In the second step, students have to familiarize themselves with the scientific manner of communicating acquired concepts in either talking or writing. This step is especially important if one considers that students need to be able to read and write mathematics when using textbooks, and be able to complete various assessment activities in writing during the course.

Whitin and Whitin [19] claim that

> Writing and talking enable learners to make their mathematical thinking visible. It is through writing and talking that teachers obtain a window into their students' thinking. Both writing and talking are tools for discovery, enabling learners to make new connections as they engage in the process. (p. 17)

Cocking and Chipman [20] referred to studies by Rosnick and Clement [21] who reported on the widespread inability of university engineering students to translate relationships expressed in natural language into corresponding mathematical expressions and vice versa. This inability can be seen as a lack of mathematical literacy. The importance of high levels of mathematical literacy becomes even more evident when one considers that fluent reading and understanding of mathematical text and symbolism are essential for studying textbooks [22].

In order to achieve the necessary in-depth mathematical understanding Thurston [16] suggested that effective communication of mathematical ideas was the key. Thurston [16] on reflecting on his career as a mathematician and how mathematics is communicated concluded that

> We mathematicians need to put far greater effort into communicating mathematical ideas. To accomplish this, we need to pay much more attention to communicating not just our definitions, theorems and proofs, but also our ways of thinking. (p. 32)

7. Mathematics and chess

Mathematics could be compared to the game of chess. To play chess well you need full memory, mastery and deep comprehension of the rules of the game and typical strategic moves. Even the brightest mind would not be a very successful chess player if before you want to move e.g. a rook, you have to spend some time recalling what the possible moves are that the rules of the game allow for a rook. This knowledge has to be intrinsic; it has to be part of your mathematical prowess.

In fact, you need more than a superficial knowledge of the rules. It is not sufficient to only know about the possible moving directions of a rook. To successfully attack or defend with your rook, you must have been exposed to repeated instances where you tried a variety of strategies attacking or defending with the rook. This intrinsic knowledge and fluency in the rules and moves of the game could be called disciplinary chess.

There are two extremes. Some chess players remain disciplinary players. To become a truly successful player, you need more than only a disciplinary knowledge of the game. Without building further on this disciplinary knowledge you will always play the game at a low level.

On the other hand, without this disciplinary knowledge, you have no hope of becoming a top class player and will easily be beaten by players with much less potential than you. Without this disciplinary knowledge it becomes impossible to develop further into the delicacies of the game.

Mathematics is not that different from a game of chess. In most topics in mathematics you need some disciplinary mathematics – the rules, basic moves and strategies of the game. Without a deep conceptual understanding of this disciplinary knowledge it is impossible to play the mathematical topic successfully. Again there is the other extreme, always playing disciplinary mathematics.

8. Examples of student problems

A big stumbling block for students is to distinguish between statements such as 'for every ε there exists a $\delta \ldots$' and 'there exists a δ for all $\varepsilon \ldots$', the latter implying there is a single δ that will 'work' for all values of ε, implying some measure of uniformity. Dubinsky and Yiparaki [12] confirm this trend. They showed that students generally prefer 'for-all-there-exists' statements to 'there-exists-for-all' statements and that most students interpret the statements as in the first form even though they were not written that way. The possible explanations the authors gave are that the first form of statement is more common in ordinary discourse or perhaps that the first form of statement is more likely to be true and students prefer true statements. These explanations are consistent with the mental model theory of deductive reasoning in the sense that it is easier to form a mental model of a familiar situation.

Often there is also a conflict between mathematical logic and logic in everyday life. Epp [23] gave an example of how 'if-then' statements are interpreted in everyday life. In mathematics, the correct negation of the statement 'if A then B' is 'A and not B', but in ordinary language there are many different interpretations for this form of statement and therefore different sensible negations. For example, when asked to negate the statement 'All mathematicians wear glasses', approximately 70% of her students selected 'No mathematicians wear glasses' instead of the correct negation 'Some mathematicians do not wear glasses'.

Statements can refer to causal relationships, temporal relationships, counter-factual relationships, etc. For example in negating a statement such as 'If I were Ann, I would not do that' we might say 'If you were Ann, you would do exactly as she did' which is not the formal negation of an 'if-then' statement [2]. Asked to negate the statement 'all girls in this class are good students', students come up with various possibilities, e.g. 'all girls in this class are poor students', or 'all boys in this class are good students', or 'all boys in this class are poor students', or even 'at least one boy in this class is a poor student'.

Another problem in everyday language use that often reflects in mathematics is the use of 'all A are not' instead of 'not all A are'. People would say 'all students in this group are not girls'. Logically this would imply that all students are boys. This is not what they mean, however, they want to say 'not all students are girls'. This reflects in mathematics as well. Students would claim that a set is not an open set if 'all of its points are not interior' rather than 'not all points are interior' or 'there is a point which is not interior'.

Marais [2] points out another ambiguity. When we work with quantified statements in everyday conversation as opposed to formal statements in mathematics, ambiguity can arise. In everyday language, the statement 'All A are B' usually implies that there exists at least one A. 'All girls in this group have blonde hair' would not be used if the entire group consists of boys. In mathematics, however, the statement will still be true even if no A exists. In her experience, students have little idea of how to start reasoning when faced with a problem with no clear answer.

Epp [23] mentions a case where students have problems when negating quantified statements. Informal ways of expressing negations of 'and' and 'or' statements in everyday discourse can lead students astray when doing negations in mathematics. For example, many students will negate '$2 \geq x \geq 1$' as '$2 < x < 1$' which does not make sense. The correct negation would be to see it as an 'and' statement and negate it to an 'or' statement '$2 < x$ or $x < 1$'. Many students do not interpret the symbol $x \geq 2$ as meaning $x > 2$ *or* $x = 2$ so they would argue that the number 3 does not qualify since it is not equal to 2.

Dubinsky and Yiparaki [12] found that students had more difficulty with determining the truth or falsity of mathematical statements than with everyday statements even though the form of the statements was the same. The explanation the authors offered for this phenomenon was that students try to build a mental picture, or mental model as in the theory of deductive reasoning, based on previous experience or imagination that represents what the statement is saying. The students therefore found the non-mathematical statements easier since it is easier to form a mental model of the non-mathematical situation, while it is difficult to form a mental model of the mathematical statements.

The data mentioned in this section seem to indicate that students do not use logic to interpret statements, but rather use mental models, which support the theory of Johnson-Laird and Byrne [6].

9. Adding structure to the transition process

If we accept the theory of mental models proposed by Johnson-Laird and Byrne [6] we see the importance of assisting students learn how to develop and use accurate mental models on the one hand, and on the other hand, they also need to be trained in the rules of logic that govern mathematical reasoning.

In an attempt to assist students to gradually develop the way of thinking needed in advanced mathematics, the transition process in developing understanding, I like to guide students in acquiring the disciplinary background in this topic, following the following strategy.

For each new concept (definition) or result (theorem) students have to be able to understand/explain the concept or result using *eight* different representations.

(I) Verbal, hand-waving, informal explanation in English (or whatever language they speak)
(II) Visually (drawing a picture)
(III) Formal mathematical symbolism
(IV) How would you begin proving that this concept is valid, or that the result is true?

The same four representations should be done for the *negation* of the concept or *converse* of the result.

Of the four different representations of a concept, numbers I and II would contribute to the concept image a student has and numbers III and IV to the concept definition. Numbers I and II would contribute to developing a mental model of the concept while numbers III and IV would contribute to the logical thinking skills of the student. Numbers I and II would be the way natural learners understand the concept while numbers III and IV would be the way formal learners approach the concept.

Negating the concept using the same four ways of expression would largely contribute to the student's logical skills in developing deeper understanding.

Expecting students to master all eight representations brings greater parity between the different thinking and learning styles – natural learners having only a concept image have to master the formal representations, the concept definitions and vice versa.

9.1. *Example*

Let us consider an example from elementary topology that would often form part of a second or third year undergraduate course in Real Analysis. To be more specific, consider the idea of an open set in a metric space. We call a set A open in a space X if for each point $a \in A$ there is a neighbourhood N around a such that $N \subset A$.
An informal way of explaining this idea is to say that all points have to be inside (interior), i.e. that the boundary is excluded. A visual picture is obvious and the formal definition is as above. With the fourth requirement, deeper understanding is required – since something has to be true for each point in A, you will choose an *arbitrary* point of A, around which you have to *construct* a neighbourhood that is still inside A. On the other hand, to begin proving that a set is not open (number eight on the list above), you will have to find (*construct*) a point that is not interior. Taking this further, a point is not interior if you cannot find a neighbourhood that is entirely inside the set, i.e. every (choose an *arbitrary*) neighbourhood around the point contains at least one point (*construct*) that lies outside the set A.

This argument clearly requires a huge cognitive leap from the student. Deeper understanding of the concept can only develop once a student has developed fluency in all eight of the mentioned ways of understanding and explaining a concept or result. Disciplinary mathematics entails the ability to employ these tools fluently, content related or non-content related.

This logical thinking process that goes with advanced mathematical thinking is experienced by students as probably the most difficult component in this transition process to conducting formal mathematics and requires fluent proficiency in their disciplinary mathematics.

The second activity that I like to involve students in is formally structured opportunities for *talking mathematics*. The value of talking mathematics was discussed earlier. Every student has to find a partner in the course. With this partner s/he gets together for one or two sessions a week. The agenda for these meetings is fixed: The two students take turns explaining the new concepts (definitions or results) to his/her partner. The speaker and the listener have equally important roles – in order to explain ideas to his/her partner, the student needs to understand the concept properly. The task of the listener is to very critically listen for mathematical errors in the explanation. Students tend to speak very loosely about mathematics. They are particularly fond of the pronoun *it*, especially when they are somewhat uncertain

about what they have in mind, e.g. 'it is a limit point', 'it converges', 'it has a maximum'. The use of the word *it* is strictly forbidden in these discussions.

With such structured guidance, students can hopefully adapt to the changing demands of advanced mathematics. It helps them to bring their concept images and concept definitions closer together. They should learn to disregard their former general way of thinking in favour of mathematical thinking. With formal instruction, this new way of thinking can become intuitive. Tall [1] views this intuition as a product of the concept images of the individual.

10. Learning mathematics

Some serious attempts have been made to develop frameworks on how students learn mathematics. Conceptual and procedural learning have been addressed by e.g. Piaget an in [24]. Gray and Tall [25,26]) introduced the idea of a *procept* and Dubinsky's [27] *APOS* theory is well known.

My own experience of the idea of *understanding* mathematics is more simplistic and does not analyse the learning process in the same detail. The process of understanding (in my experience) is an ongoing process converging to full understanding but never reaches the limit of full understanding. In this model, the dynamic process of understanding new mathematics takes place in layers. With every layer you understand a little deeper. Some people have the capacity to understand in thick layers. The mathematics community considers these students as bright. They understand quickly. Unfortunately in many instances some of these people digging in thick layers sometimes falsely think that they understand fully – which is unlikely. This has as consequence that they do not consider it necessary to visit the topic again.

Often, the not so bright mathematics students, realizing that their layers of new understanding are thinner, deliberately go for repeated exposure to understand deeper. With this repeated exposure, going through a next (thin) layer every time, some of these students' understanding grows and they eventually understand deeper than some of the bright students. To some extent it is sometimes better to be not too bright – in the end you may have deeper conceptual insight in mathematics and are better equipped to discover/develop new mathematics. This is, of course, if you have the perseverance to expose yourself repeatedly to the particular concept. In my personal experience: I have taught courses in multivariate calculus many years and have to confess that every year that I teach this course, I learn something new. Something that I suddenly realize that I had never really understood properly. Something that makes me realize I am still digging deeper into understanding completely. My layers are getting thinner every year – it feels like a limiting process with definite convergence. But I realize I will never get there. The progress of understanding never settles. Perhaps not so much because one is making more logical inferences but also because one is always finding new images, making new connections, seeing things in broader contexts.

This process of understanding is also misleading. Mathematics can be the easiest subject in the world if you watch somebody else do it. A good teacher explains difficult mathematics so well that you are convinced that you understand everything – s/he makes it look so simple. It is only when you try to do it yourself that you

suddenly are confronted by the intricacies, the cognitive depth of a concept. It is only then that you realize how thin your first layer of understanding really was.

Rote, repeated exposure to mathematical problems have been condemned by educationists over the last few decades, in favour of a constructive, experimental approach in which every student has to discover or construct new concepts and develop his/her own algorithms. I share this view only partially. In my opinion, learning mathematics consists of two processes, the first-time exposure and the consolidation process. Granted, in the first-time exposure to a new idea or concept it is much more pleasing and sensible to discover or construct it yourself than when you are merely informed by the teacher and this approach will surely enhance understanding and longer-term retention. However, deeper understanding can only grow with repeated exposure. This implies that drill-and-practice exercises do have a place – doing more problems of a certain type rather than only one brings repeated exposure and deeper eventual understanding. Being exposed to the moves of mathematics repeatedly, one acquires the disciplinary mathematics required for conducting mathematics on a deeper level – similar to chess.

For formal learners – as most of the students in our classes are – moving from the type of mathematical activity that they are familiar with to advanced mathematical activity does not appear natural. These students have to be assisted in this transition process in a structured manner and by repeated exposure to the new ideas of advanced mathematics. The process of moving to better understanding of the concepts, ideas and ways of thinking in advanced mathematics can be followed successfully with structured, rote-type of activities, such as those suggested in this article.

Repeated structured exposure as suggested here should help students' understanding of mathematical concepts grow. Most important is that they engage in this activity themselves. By doing their thinking for them as a teacher, students may find it difficult to move from general thinking to mathematical thinking.

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